

Anto Idicherian Lonappan

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Academic Appointments

- 2024 – present ♦ **Postdoctoral Research Fellow**, University of California, San Diego.
PI: Prof. Brian Keating
- 2022 – 2024 ♦ **Postdoctoral Research Fellow**, University of Rome, Tor Vergata.
PI: Prof. Nicola Vittorio

Education

- 2018 – 2022 ♦ **Ph.D in Astrophysics and Cosmology.**
Scuola Internazionale Superiore di Studi Avanzati-SISSA
Thesis title: *Cosmic Microwave Background: Analysis and Interpretation of B Modes for the detection of Inflationary Gravitational Waves and Gravitational Lensing.*
Awarded Cum Laude (with honors)
Advisor: Prof. Carlo Baccigalupi
- 2014 – 2016 ♦ **M.Sc. Physics**
St. Berchmans College, Mahatma Gandhi University
Thesis title: *Markov Chain Monte Carlo Methods: Cosmological Parameter Estimation*
Advisor: Dr. Charles Jose
- 2010 – 2013 ♦ **B.Sc. Physics**
St. Berchmans College, Mahatma Gandhi University

Research Experience

- 2022 ♦ **Visiting Research Fellow**, Kavli-IPMU University of Tokyo, Japan.
- 2017 – 2018 ♦ **Junior Research Fellow**, Presidency University, India.
- 2016 – 2017 ♦ **Research student**, Centre for Theoretical Physics, JMI, India.

Memberships

- ♦ Astronomical Society of India

Scientific Collaborations

- CMB ♦ POLARBEAR, Simons Array, Simons Observatory, LiteBIRD, CMB-S4
- GW ♦ LISA
- Others ♦ CosmoVerse, CMBAgent

Referee

- ◇ Physical Review Letters (PRL)
- ◇ Physical Review D (PRD)
- ◇ Astronomy & Astrophysics (A&A)
- ◇ Journal of Cosmology and Astroparticle Physics (JCAP)

Awards & Grants

- ◇ NERSC AI4Sci CFP - 2000 GPU Hrs for the proposal titled "Inflationary B -mode detection and measurement for the Simons Observatory using spherical convolutional neural networks"
- ◇ NERSC AI4Sci CFP - 10000 CPU Hrs for the proposal titled "Hybrid Classical-Neural Foreground Cleaning for Next-Generation CMB Polarization Experiments"
- ◇ H2020 - Marie Skłodowska-Curie Research and Innovation Staff Exchange - 2020 European grant.
- ◇ Junior Research Fellowship(2017)-Department of Science and Technology, Government of India.

Selected Invited Talks

- ◇ Testing parity-violating Physics with CMB.
University of Southern California, LA, USA.
- ◇ Multi-Tracer Delensing strategy for LiteBIRD.
CMB-Bharat Seminar, India.
- ◇ Probe combination with CMB-S4 and LiteBIRD to improve sensitivity on Cosmic Inflation
IPMU-University of Tokyo, Japan
- ◇ Gravitational Lensing of CMB: Scope of LiteBIRD
University of Rome, Tor Vergata, Italy.
- ◇ Cosmic Birefringence And Simons Observatory Calibration.
University of Mila, Italy

Services and Leadership

- ◇ Remote Observation Coordinator, Simons Observatory.
Three Small Aperture Telescope and a Large Aperture Telescope
- ◇ Builders Status Panel, Simons Observatory.
- ◇ Junior Member Organization, Simons Observatory.
- ◇ Physics Area representative, Students council, SISSA.

Certificates

- ◇ Deep Learning and Reinforcement Learning
- ◇ Unsupervised Machine Learning
- ◇ Supervised Machine Learning: Classification
- ◇ Supervised Machine Learning: Regression
- ◇ Introduction to Computer Science and Programming Using Python

Teaching

- ◇ Introduction to CMB.
<https://antolonappan.me/tutorial/cmb/>

Skills

- Coding ◇ PYTHON, C, C++, R, MPI, Bash, Zsh, $\LaTeX$
- Databases ◇ MySQL, SQLite.
- Web Dev ◇ HTML, CSS, JavaScript, Apache, Nginx.
- OS ◇ LINUX and MacOS.
- Prompt Eng. ◇ OpenAI, Claude, Gemini.

Publications

Selected Articles

- 1 Lonappan, A. I., Ramesh, K., Namikawa, T., Qu, F. J., & Keating, B. (2026). *Constraints on cosmic strings from the curl-mode cmb lensing power spectrum measured by act dr6*. arXiv: 2606.19337
- 2 Ruiz-Granda, M. et al. (2025). *LiteBIRD Science Goals and Forecasts: Improved full-sky reconstruction of the gravitational lensing potential through the combination of Planck and LiteBIRD data*. arXiv: 2507.22618 [astro-ph.CO]
- 3 Xu, L., Sarkar, M., Lonappan, A. I., Zubeldia, Í., Villanueva-Domingo, P., Casas, S., ... Bolliet, B. (2025). *Open Source Planning & Control System with Language Agents for Autonomous Scientific Discovery*. [doi:10.48550/arXiv.2507.07257](https://doi.org/10.48550/arXiv.2507.07257). arXiv: 2507.07257 [cs.AI]
- 4 Lonappan, A. I. (2025). *Improving cosmic birefringence constraints via delensing*. [doi:10.1088/1475-7516/2025/07/009](https://doi.org/10.1088/1475-7516/2025/07/009). arXiv: 2503.04708 [astro-ph.CO]
- 5 Lonappan, A. I., Keating, B., & Arnold, K. (2025). *Constraints on anisotropic cosmic birefringence from CMB B-mode polarization*. [doi:10.1103/yxmh-rh9z](https://doi.org/10.1103/yxmh-rh9z). arXiv: 2504.13154 [astro-ph.CO]
- 6 Lonappan, A. I. et al. (2024). *LiteBIRD science goals and forecasts: a full-sky measurement of gravitational lensing of the CMB*. [doi:10.1088/1475-7516/2024/06/009](https://doi.org/10.1088/1475-7516/2024/06/009). arXiv: 2312.05184 [astro-ph.CO]
- 7 Namikawa, T., Lonappan, A. I. et al. (2024). *LiteBIRD science goals and forecasts: improving sensitivity to inflationary gravitational waves with multitracer delensing*. [doi:10.1088/1475-7516/2024/06/010](https://doi.org/10.1088/1475-7516/2024/06/010). arXiv: 2312.05194 [astro-ph.CO]
- 8 Adachi, S. et al. (2022). *Improved Upper Limit on Degree-scale CMB B-mode Polarization Power from the 670 Square-degree POLARBEAR Survey*. [doi:10.3847/1538-4357/ac6809](https://doi.org/10.3847/1538-4357/ac6809). arXiv: 2203.02495 [astro-ph.CO]
- 9 Chowdhury, R. K., Chatterjee, S., Lonappan, A. I., Khandai, N., & DiMatteo, T. (2019). *Cosmological Simulation of Galaxy Groups and Clusters-I: Global Effect of Feedback from Active Galactic Nuclei*. [doi:10.3847/1538-4357/ab5b96](https://doi.org/10.3847/1538-4357/ab5b96). arXiv: 1911.07824 [astro-ph.GA]
- 10 Lonappan, A. I., Kumar, S., Ruchika, Dinda, B. R., & Sen, A. A. (2018). *Bayesian evidences for dark energy models in light of current observational data*. [doi:10.1103/PhysRevD.97.043524](https://doi.org/10.1103/PhysRevD.97.043524). arXiv: 1707.00603 [astro-ph.CO]